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Prevalence of multidrug-resistant bacteria associated with polymicrobial infections

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ABSTRACT

Background: Wounds remain the most important cause of postoperative mortality and morbidity and generate considerable additional social and healthcare costs. Most wounds are caused by various coliforms, *Enterococcus fecalis*, *Proteus* sp., and multidrug resistant *Staphylococcus aureus*. Wound is one of the leading cause of infections in the under developed and developing countries than developed nations.
Methods: A total of 43 samples associated with bacteremia and wound infection were collected. Biochemical characterization and culture characteristics of the drug resistant isolates were studied using MacConkey agar, blood agar and mannitol-salt agar. Antibiotic susceptibility analysis of the isolated strains was performed by disc diffusion method using various antibiotics. Prevalence of dug resistance among bacteria isolated from the wound was studied. The ability of Beta lactamase antibiotic producing bacterial strains were analyzed.
Results: A total of 168 bacterial strains were isolated showed high resistant towards ampicillin (89%), ciprofloxacin (90.8), cefepine (90.5), piperacillin (91.8), oxacillin (92.5), and imipenem (96.5). The isolated bacterial strains showed monobacterial as well as polybacterial growth on the surface of the wound. The isolated bacterial strains revealed 89% sensitivity against norfloxacin and 94.9 sensitivity against vancomycin. About 26% of bacterial strains degraded quinolones, whereas only 14% clinical isolates showed their ability to degrade aminoglycosides. A total of 27% bacteria degraded tetracycline and 51% of isolates degraded carbapenems compounds. Interestingly, *E. faecalis* was resistant against antibiotics such as, Oxacillin, Nalidic acid, Ofloxacin, Erythromycin, Norfloxacin, Ciprofloxacin, Ampicillin, Tetracycline, Cefepine, Amikacin, Cefurooxime, Vancomycin, Piperacillin, Imipenem and Gentamycin. Moreover, *Proteus* species was resistant against certain numbers of antibiotics namely, Ampicillin, Piperacillin, Oxacillin, Nalidic acid, Tetracycline, Erythromycin, Cefurooxime, Nitrofurantoin, Vancomycin and Imipenem.
Conclusions: The isolated bacterial strains were resistant against various drugs including vancomycin. Staphylococci, and *E. faecalis*is strains showed resistance against various classes of antibiotics.

Introduction

Antimicrobial resistance (AMR) is an evolutionary response of bacteria, fungi and viruses to withstand commercial antimicrobial drugs [1]. These bacteria showed resistant towards various antibiotics due to their activity of efflux pumps, cell wall structure or presence of porins. The resistant bacterial strains within the species are resistance to the provided antimicrobial drugs. The drug resistance mechanisms include either severe mutations in existing genes, for example in core metabolic genes, intracellular targets, or acquisition of antibiotic resistance genes through horizontal gene transfer. Horizontal gene transfer enables inter- and intra-specific transmission of various antimicrobial resistance genes and is highly responsible for an antimicrobial resistance pandemic [2]. Antimicrobial resistance existed earlier before the

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invention of antibiotics and unrestricted application of antibiotics in animal husbandry, poor wastewater management, overuse of antibiotics in the hospitals and poor and low income countries has led to wide spread of drug resistant bacteria throughout the world [3]. The worrisome spread of drug resistance genes is colistin-resistance genes in *Pseudomonas aeruginosa*, *Acinetobacter baumannii* and *Enterobacteriaceae*; plasmid mediated spread of carbapenemases [4], methicillin resistance gene in *Staphylococcus aureus*, vancomycin resistance gene in *Staphylococcus aureus* and *Enterococci* [5].

Health-care-associated infections are associated with health care centres or acquired from hospitals. These health-cares associated infections may occur after 28 h of hospitalization or after discharge of patients [6]. Many findings are available on drug resistant bacterial strains in the hospital environment [7,8]. The bacteria such as, *Pseudomonas* sp., E*nterobacter* sp., *Streptococcus* sp., *Proteus* sp., *E. coli*, *Kelbsiella* sp. and *S. aureus* and Staphylococci strains have been reported as the common bacterial pathogens [9,10]. *P. aeruginosa* is one of the opportunistic nosocomial pathogens, which involves a broad spectrum of various infections and leads to considerable morbidity in most of the immune compromised patients. Due to drug resistance to various available antibiotics, the mortality rate is considerably high [11]. Most of the clinical strains are drug resistant, including vancomycin and macrolide. These bacterial strains have the ability to produce β-lactamases, causing drug resistance to cephalosporins and penicillins [12,13]. Recently few studies have focused on screening of *E. faecalis* from bacteremia and surgical wound infections. However, little information is known about drug resistance mechanism, antibiotic susceptibility bacteria associated with surgical wound. Moreover, the epidemiological analysis clearly indicates the different nature of these bacterial infections and the incidence varies between regions, wound classes and procedure. In recent years, the emergence of drug resistance, multiple drug resistant strains of various hospital acquired bacterial pathogens become a challenge [14]. However, the additional use of antimicrobial, readmissions, treatment failure, and increase of hospital stay was expected for admitted patients infected by multiple drug resistant strains. The rise of multidrug resistant Gram-negative bacterial pathogens is of important concern since severing mortality risk of various infections including nosocominal infections caused by multidrug resistant bacteria than non-multidrug resistant bacteria [15]. Surveillance results about antimicrobial resistance and etiology is an important way to overcome the risk associated with treatment failure and drug resistance. Hence, the present investigation aimed to analyze the prevalence of drug resistant bacteria in wound and antibiotic resistance of frequently determined clinical bacterial strains.

Materials and methods

Characterization of bacterial strains

Wound swabs were collected and process within two hours of sample collection. Two swabs were made from a samples. One swab was utilized for the preparation of biochemical characterization and other swab was utilized for the inoculation into MacConkey agar, Blood agar and mannitol-salt agar. The plates were incubated for 48 h at 37 °C. Gram-negative bacterial strains were characterized based on morphological features on MacConkey agar and blood agar, followed by biochemical experiments such as, urease test, citrate test, indole and motility test, triple sugar iron test and oxidase test.

Table 1
Antibacterial susceptibility of the isolated bacteria.

Antibiotics	Antibiotic susceptibility	
	Sensitive (%)	Resistant (%)
Ampicillin	11	89
Nalidic acid	23	77
Tetracycline	52	48
Norfloxacin	89	11
Amikacin	55.5	44.5
Ciprofloxacin	9.2	90.8
Cefurooxime	15.5	84.5
Nitrofurantoin	11.5	88.5
Cefepine	9.5	90.5
Vancomycin	94.9	5.1
Ofloxacin	30.5	69.5
Piperacillin	8.2	91.8
Oxacillin	7.5	92.5
Imipenem	3.5	96.5
Erythromycin	50.9	49.1
Gentamycin	28.1	71.9

Antibiotic susceptibility test

Antibiotic susceptibility analysis of the isolated strains was performed by disc diffusion method. Antibiotics such as, Nitrofurantoin, Oxacillin, Nalidic acid, Ofloxacin, Erythromycin, Norfloxacin, Ciprofloxacin, Ampicillin, Tetracycline, Cefepine, Amikacin, Cefurooxime, Vancomycin, Piperacillin, Imipenem and Gentamycin were used. Mueller-Hinton agar medium has been used for the determination of antibiotic susceptibility test. After 24 h incubation, drug resistance properties were studied.

Beta lactamase production by the bacterial strains

In this experiment, clavulanic acid/amoxicillin disc (Himedia, Mumbai, India) was placed in the Petri plate containing Mueller-Hinton agar. To this plate predetermined quantities of culture was swabbed on it previously. The antibiotics such as, ceftriaxone, ceftazidime and cefotaxime were place on MHA plates. Zone of inhibition around the colony was observed after 24 h incubation and beta lactamase producing property was analyzed.

Results

Antibiotic susceptibility of the bacterial strains

The isolated strains showed high resistant on ampicillin (89%), ciprofloxacin (90.8), cefepine (90.5), piperacillin (91.8), oxacillin (92.5), and imipenem (96.5). The isolated bacterial strains were monobacterial nature as well as polybacterial growth. These bacterial strains showed 89% sensitivity against norfloxacin and 94.9 activities against vancomycin (Table 1).

Beta lactamase producing pathogenic bacterial strains isolated from wound infection

The antibiotics or its analogue was used for the determination of beta lactamase by the clinical isolates. The antibiotics such as, quinolones, tetracycline, aminoglycosides and carbapenems were used to test degrading ability of the strains (Table 2). About 26% of bacterial strains degraded quinolones, whereas only 14% clinical isolates showed their ability to degrade most of the aminoglycosides. A total of 27% bacteria degraded tetracycline and 51% of isolates degraded carbapenems compounds.

Table 2
Drug degrading ability of the bacterial strains isolated from the wound.

Antibiotics	Beta lactamase production	
	Non producer (n and %)	Producer (n and %)
Quinolones	40 (64)	26 (36)
Tetracycline	36 (57)	27 (43)
Aminoglycosides	49 (73.5)	14 (6.5)
Carbapenems	12 (19)	51 (81)

Table 3
Prevalence of bacterial species in the surgical wound of patients.

NO	Bacteria	Availability (%)
1	*Citrobacter* spp.	13.2
2	*S. aureus*	14.2
3	*Proteus* spp.	25.9
4	*P. aeruginosa*	5.2
5	*E. coli*	10.2
6	*K. pneumoniae*	9.2
7	*E. faecalis*	18.2
8	*Enterobacter* spp.	3.9

Prevalence of bacteria

A total of eight bacterial genera were characterized in this study. The predominant bacteria were identified from the genus, *Proteus* and *Enterococcus faecalis*, *Proteus* spp., followed by *S. aureus*, and *Citrobacter* spp. These bacterial strains were either from monomicrobial or polymicrobial nature. All these detected bacteria were cultured in various selective media and sub cultured for every two months. The most predominant bacterial genera were *Proteus* spp. (25.9%) and *Enterococcus* spp. (18.2%). The prevalence of bacteria and availability was described in Table 3.

Growth of clinical isolates on MacConkey agar

Enterococcus faecalis and *Proteus* sp. were predominant in surgical wound of the patients. On MacConkey agar medium, *E. faecalis* strains showed their ability to ferment lactose, whereas *Proteus* did not show any results and the plates remains same without any colouration (Fig. 1). *Proteus* sp. appeared as slimy layer on agar plates because of the production of exopolysaccharides.

Antibiotic susceptibility of abundant bacteria

Antibiotic susceptibility test was performed with the pure culture of *Enterococcus faecalis* and *Proteus* sp. *E. faecalis* showed resistance against most of the tested antibiotics. *Proteus* species also showed drug resistant properties, and also has multidrug resistance properties. Antibiotic resistance properties of selected multidrug resistant strains were described in Fig. 2. *E. faecalis* was resistant against antibiotics such as, Oxacillin, Nalidic acid, Ofloxacin, Erythromycin, Norfloxacin, Ciprofloxacin, Ampicillin, Tetracycline, Cefepine, Amikacin, Cefurooxime, Vancomycin, Piperacillin, Imipenem and Gentamycin. It showed sensitivity against Nitrofurantoin. Moreover, *Proteus* species was resistant against certain numbers of antibiotics namely, Ampicillin, Piperacillin, Oxacillin, Nalidic acid, Tetracycline, Erythromycin, Cefurooxime, Nitrofurantoin, Vancomycin and Imipenem (Fig. 2; Table 4).

Discussion

The abundance of drug resistant bacterial strains was screened from the clinical specimens. The bacterial strains such as, *Klebsiella* sp., *Pseudomonas* sp., *Enterobacter* sp., and *E. coli* strains were predominant among individuals. Wound infection rate varied based on patients and hospitals as well as sterile condition of the hospital environment. Generally, in developed and most of the developing countries, the wound infection rate is low than poor countries. In Nigeria 74.9% wound infection cases have been reported [16] and North Ethiopia 44.1% cases were associated with wound infection [11]. The clear variation in prevalence of post-operative infection may be difference in study design used by the research groups, hospitals, prevention between countries, variation in policy of infection control, and variation in common nosocomial pathogens inhabitant. Multiple infections in patients in surgical ward have been reported previously. *E, faecalis* and *Proteus* sp. were the two most common bacteria characterized in this study. Other than these two strains, *Klebsiella* sp., and multidrug resistant *S. aureus* also observed in this study which was in agreement with previous finding. Most of the bacterial strains detected in this study were abundant in the skin. These bacterial strains easily enter in to the surgical site during hospital stay.

Enterococci have been predominant among hospitalized patients and these bacteria were generally reported as the one of the most important hospital-acquired pathogens [17]. These bacterial species have been isolated from various surgical wounds and

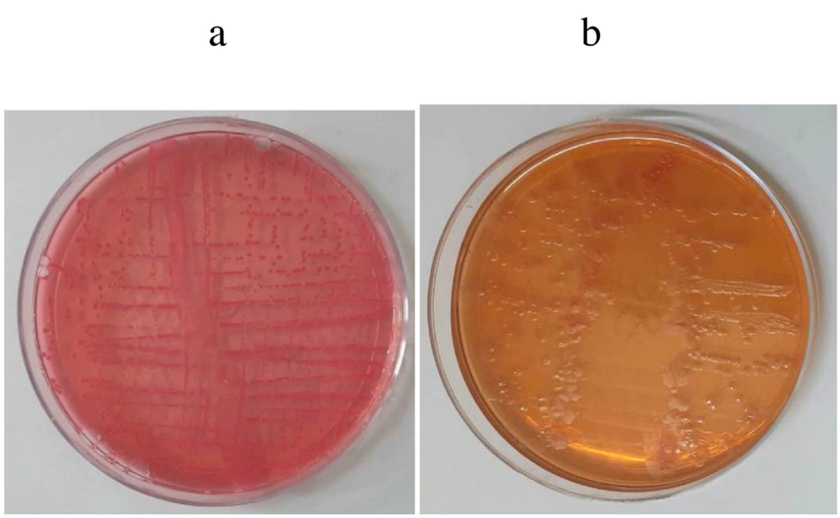

Fig. 1. Growth of clinical bacterial strains on MacConkey agar medium. Growth of *E. faecalis* on MacConkey agar medium appeared as red due to lactose fermentation (a), and no colouration was obtained in the medium cultivated with *Preteus* sp. becasue of inability to ferment lactose.

a b

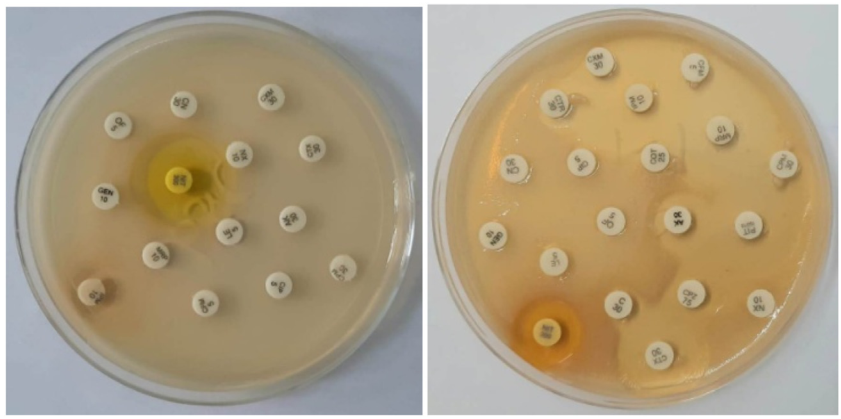

Fig. 2. Antibiotic susceptibility of the selected bacterial strains. Antibiotic susceptibility of *E. faecalis* (a) and *Proteus* sp.

Table 4
Drug resistance pattern of clinical isolates.

Antibiotics	Antibiotic susceptibility	
	E. faecalis	*Proteus* sp.
Ampicillin	R	R
Piperacillin	R	R
Oxacillin	R	R
Norfloxacin	R	S
Amikacin	R	S
Nalidic acid	R	R
Tetracycline	R	R
Erythromycin	R	R
Gentamycin	R	S
Ciprofloxacin	R	S
Cefurooxime	R	R
Nitrofurantoin	S	R
Vancomycin	R	R
Cefepine	R	S
Ofloxacin	R	S
Imipenem	R	R
Cephalothin	R	S
Cefotaxime	R	S

traumatic wounds. The availability of these Enterococci population varied widely. In this study two most predominant bacterial strains were subjected for antibiotic susceptibility test. The bacterial population observed in this study was similar with findings from earlier studies performed in other countries. In wounds, both Gram-positive and Gram-negative bacterial strains dominated considerably. Previous studies both from Europe, Africa and U.S confirm *P.aeruginosa* and *S. aureus* as the leading microbial pathogens isolated from wounds. Both bacterial strains show surface proteins and virulence factors negatively influencing wound healing properties [18]. In poly-and monomicrobial infections, P. aeruginosa was considered as the dominating bacterial strains. The bacterial species from enterococci and Enterobacteriaceae have been rarely observed alone, and it is probable they are minor components or contamination flora of polymicrobial infection. The development of antibiotic resistance among bacteria poses serious risk in both pediatric and adult patients. The factors such as prolonged and excessive use of antibiotics, antibiotic therapy without knowledge of circulating bacterial strains, and hospital hygienic practices have lead to the selection and development of multidrug resistant bacteria [19]. Carbapenem is generally considered the last line antimicrobial agent to treat patients associated with multidrug resistant bacterial strains. The increased antibiotic resistance among the Gram-negative bacteria (non-fermentative type), namely, *Acinetobacter* spp., *Acinetobacter* spp., and *P. aeruginosa* have been resistance against various antibiotics. Moreover, the bacterial strains have been showed very less resistance to antibiotics such as, meropenem and imipenem [20]. Similarly, bacterial strains from *P. aeruginosa* have been showed drug resistance against various third-generation antibiotics such as, aztreonam, ciprofloxacin, piperacillin, and cephalosporins [21,22]. Erythromycin resistant bacteria have been previously isolated from wound. Surgical site infections and surgical wounds with inflammatory tissues are largely polymicrobial in nature and the role of both anaerobic and aerobic bacteria in the pathogenesis of these nosocomial infections is recognized. Gram-negative bacteria were predominant in wounds than Gram-positive because most of the bacteria were Enterobacteriacea microflora. Multidrug resistant bacteria were defined as a bacterial isolate with resistance to at least three or more antibiotics [23]. Polymicrobial habitat of these bacterial infections and persistence of anaerobic bacterial strains in surgical site infections can cause failure in antibacterial treatment. The emergence of extensive spectrum β-lactamase-producing ability of the isolated bacterial strains was determined. B-lactamase-producing multi drug-resistant *E. coli*, *P. aeruginosa*, *K. pneumoniae* and *E. faecalis* have become a serious problem in under developed and developing countries. From the present study it was found that most of the bacterial strains from the strains *E. faecalis* were resistant against various drugs than *Proteus* species. These bacterial strains synthesized β-lactamases and most prevalent drug resistant strains were found in folate pathway inhibitor of antimicrobial classes and tetracycline class. The high incidence of antibacterial resistance revealed unrestricted use of these antibiotics. Ullah et al. [24] studied a high incidence of folate pathway inhibitor resistance and tetracyclines resistance. Pathogenic bacteria belong to various β-lactams class, including carbapenems resistant isolates, and β-lactam inhibitors resistant isolates. Altoparlak et al. reported that 75% of studied pathogenic bacteria were high resistance to cephalosporins. Antibacterial resistance in enterococci group of bacteria is mainly due to penicillin-binding protein modification and betalactamase production. In bacteria, resistance by lactamases was mainly due to resistance to gentamicin. Most of the identified *Enterococcus* bacteria have the ability to resistance to lincosamides [25–28]. Previous studies reported that various bacteria from Enterococcus showed maximum resistance to aminoglycosides. Oxazolidinones class of antibacterial agents showed activity against bacteria, including *Enterococcus faecalis*, resistant or susceptible to glycopeptides. The structure of extensive spectrum beta lactamase has changed in last two decades. In recent years, the spread of third generation cephalosporinshas been reported in various countries [29].

Nosocominal infections can be mainly endogenous origin and patients own microbial flora is an important source of infection. According to WHO, about 8.7% of patients associated with nosocominal infections. In some cases pathogenic bacteria come in contact from surfaces, air, water or in the hospital environment, staff and other in or out patients [30–32]. In recent years, multidrug resistant bacterial strains, namely, Pseudomonas aeruginosa, Acinetobacter baumannii, glycopeptide-resistant Enterococci, and methicillin-resistant *Staphylococcus aureus* were reported. Manual contact between patients and care giver was one of the areas of disease transmission. This is also one of the non-negligible modes of pathogenic bacteria transmission manually among infected patients, care giver and environment. Among these bacteria, 15% were Gram-negative bacilli, 20% were GRE, and 30% of which were MRSA. The isolated *E. faecalis* strains showed resistant on oxacillin and vancomycin. The incidence of vancomycin resistant Staphylococci in hospital is alarming because this antibiotic is currently using as the main antibacterial agents to treat multidrug resistant bacteria including MRSA. Unlike *E. faecalis*, *Proteus* was sensitive to various common antibiotics used for analysis. The high antibiotic susceptibility of bacterial species has been previously [33]. In recent years, antibacterial secondary metabolites have been reported for the treatment of various multidrug resistant bacteria isolated from various sources [34–38]. Analysis of drug resistance pattern from clinical samples is useful for the development of novel drugs.

Conclusions

Antibiotic resistance increases risk throughout the world. The increased proportion of multidrug resistant bacteria determined in this study was alarming because of lack of suitable treatment options for the infections. The isolated bacterial strains were resistant against various drugs including vancomycin. The incidence of vancomycin resistant in bacteria associated with infection is increasing in recent years. *E. faecalis*is is one of the bacterial isolates showed resistance against various classes of antibiotics.

Conflict of interest statement

None declared.

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